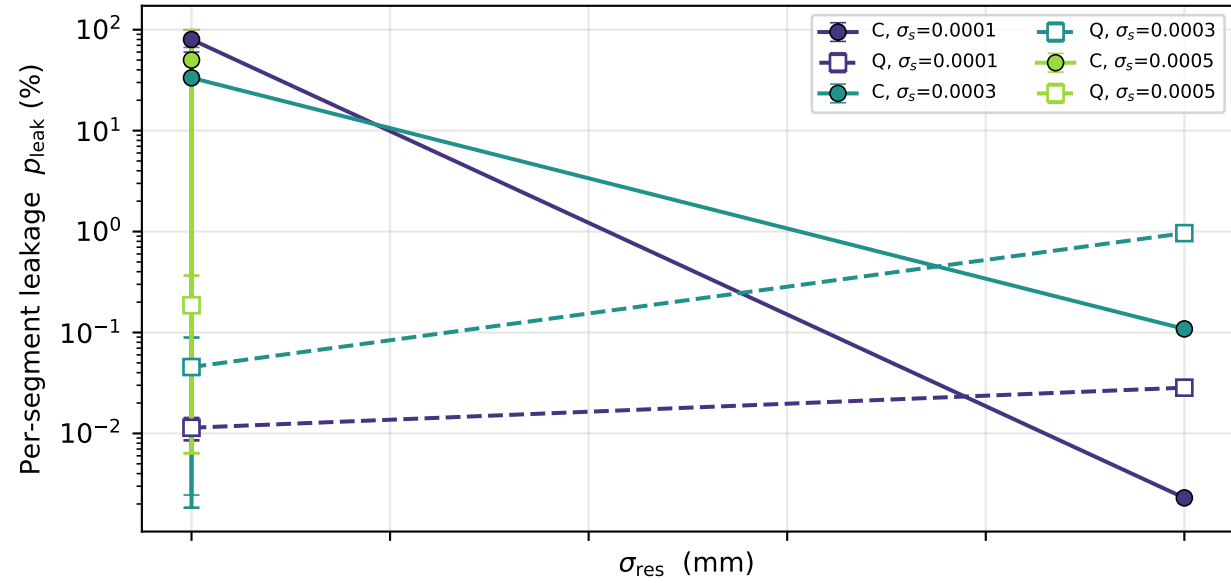
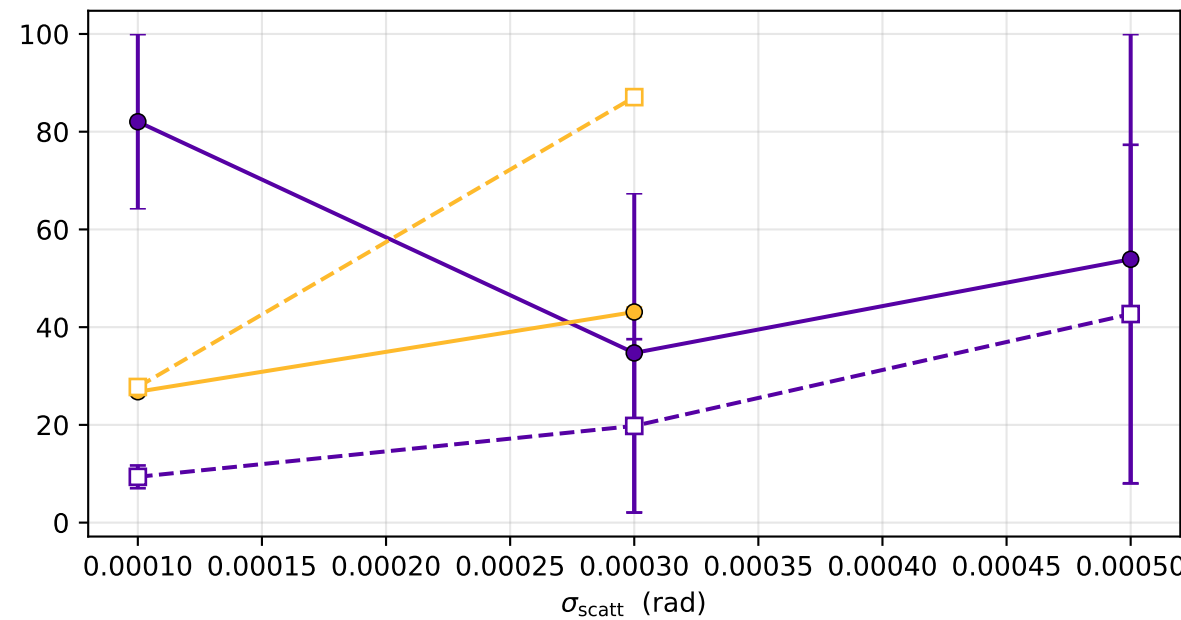
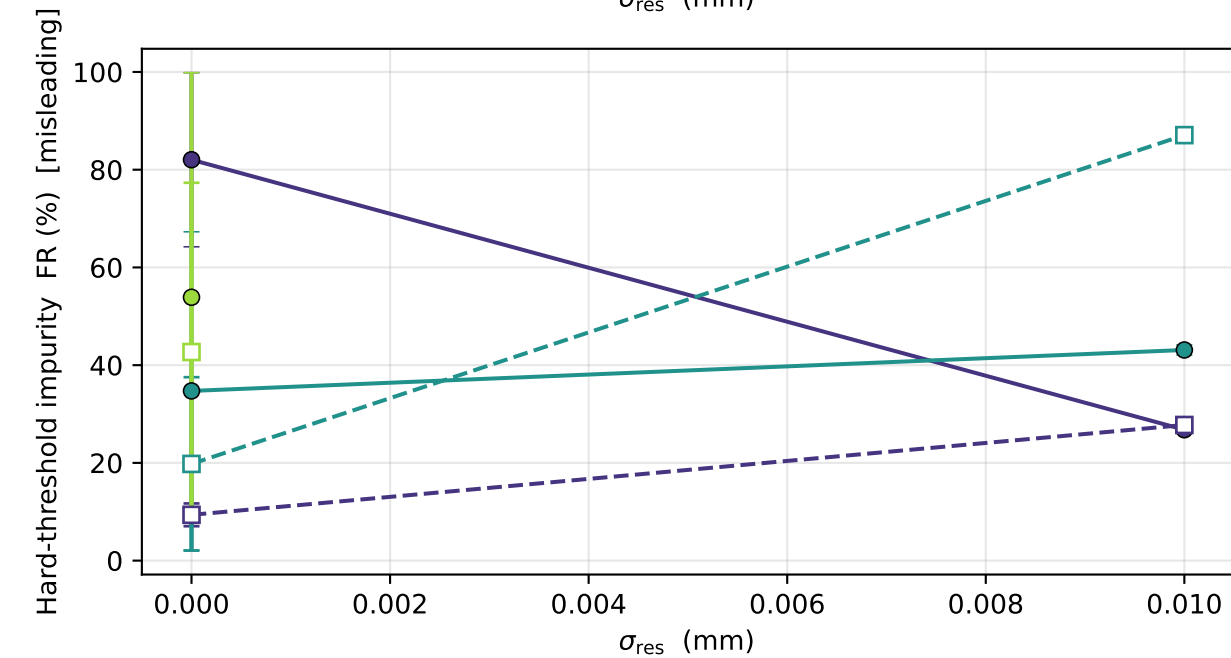
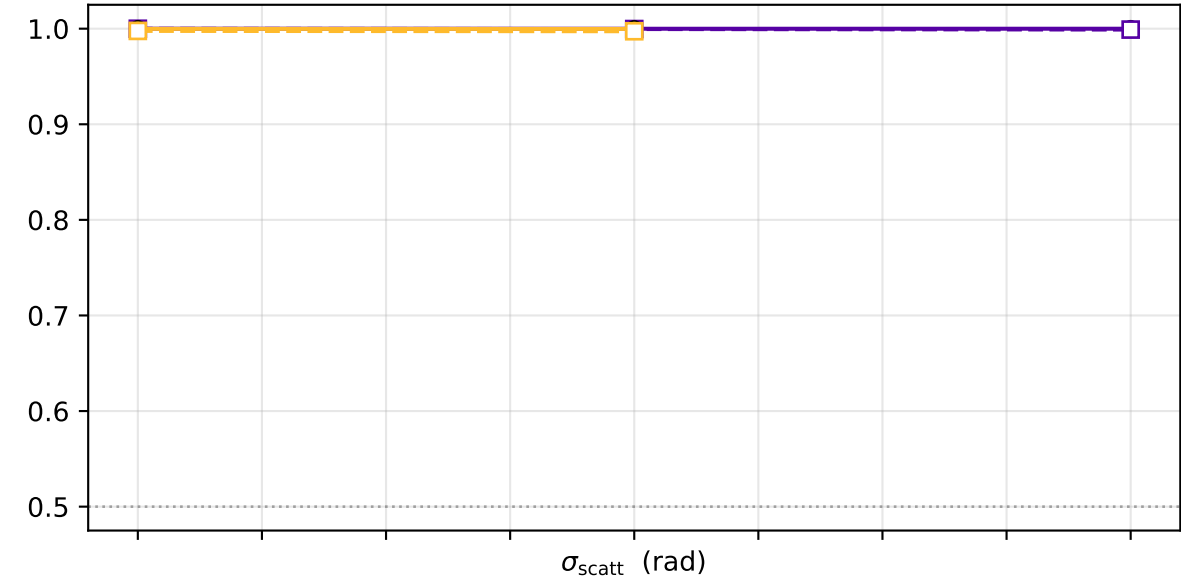
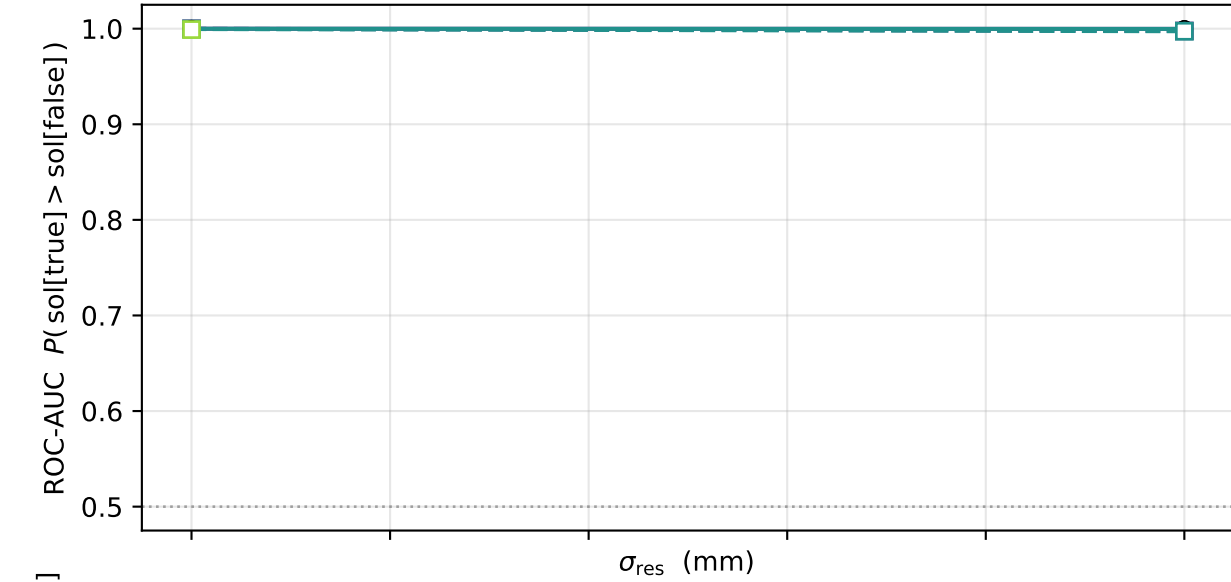
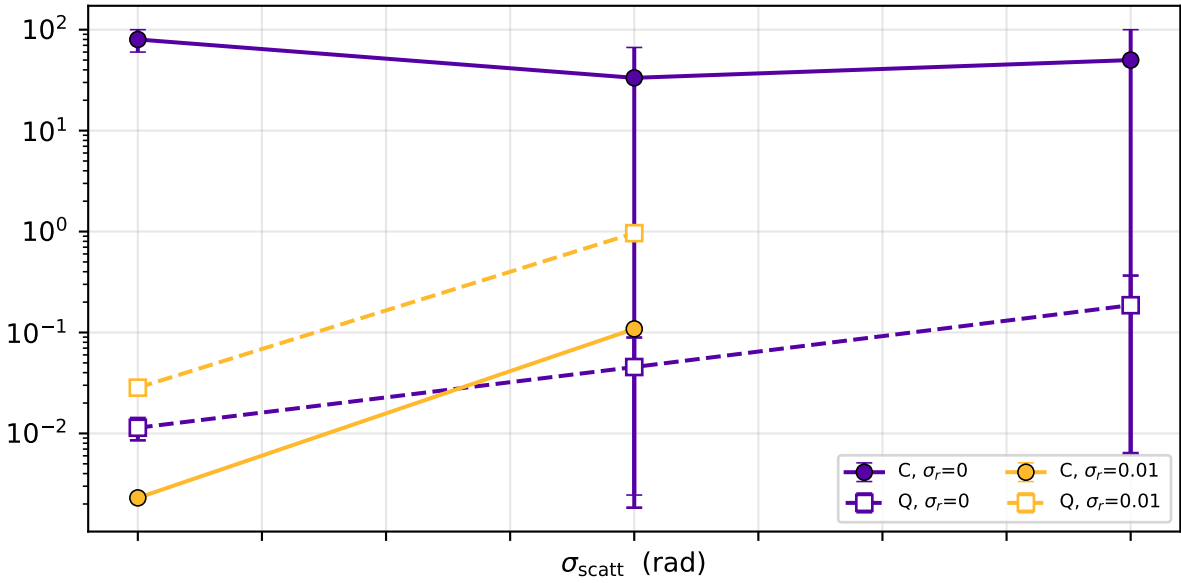


Threshold-robust segment metrics — T = 700 tracks (solid = classical, dashed = 1-bit-HHL)

vs hit resolution



vs multiple scattering



Controls: $\gamma=3, \delta=1, \phi_{\text{max}}=\theta_{\text{max}}=0.2 \text{ rad}, \theta_{\text{min}}=1.5\text{e-}5, \tau=0.35, \varepsilon=\text{calc.}(s=3), \text{conv OFF, hit-ineff 0, 5 planes } \Delta z=33 \text{ mm, statevector 1BQF}$
 Fixed here: per-segment leakage & impurity at $\tau=0.35$; AUC threshold-free
 Swept: $\sigma_{\text{res}}, \sigma_{\text{scatt}}, T$